

3. Researching a topic, evaluating information, and Literature survey

Literature survey

A literature survey or a literature review in a project report is that section which shows the various analyses and research made in the field of your interest and the results already published, taking into account the various parameters of the project and the extent of the project.

When you write a literature review in respect of your project, you have to write the researches made by various analysts - their methodology (which is basically their abstract) and the conclusions they have arrived at. You should also give an account of how this research has influenced your project.

It is the most important part of your report as it gives you a direction in the area of your research. It helps you set a goal for your analysis - thus giving you your problem statement.

A literature review must contain at least 5 - 7 published researches in your field of interest.

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Arduino - Home

<https://www.arduino.cc/> ▾ Arduino ▾

Open-source electronic prototyping platform enabling users to create interactive electronic objects.

Results from arduino.cc



Arduino - Software

The open-source Arduino Software (IDE) makes it easy to ...

Introduction

What is Arduino? Arduino is an open-source electronics ...

Arduino - Products

Browse the full range of official Arduino products, including ...

Arduino Uno

Arduino/Genuino Uno is a microcontroller board based on ...

Arduino (@arduino) | Twitter

<https://twitter.com/arduino>

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The AtHeart 4Duino combines Arduino, Wi-Fi and touchscreen: bit.ly/2dtbaJg
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It's Ada Lovelace Day, a day to celebrate the achievements of women in STEM!
#AdaLovelaceDay
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Arduino - Wikipedia, the free encyclopedia

<https://en.wikipedia.org/wiki/Arduino> ▾ Wikipedia ▾

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power transistor



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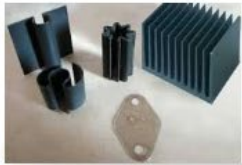
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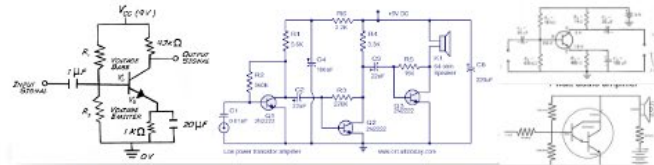
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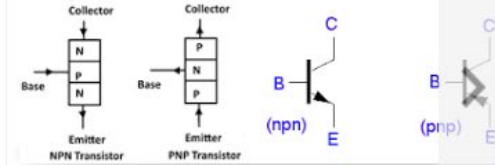
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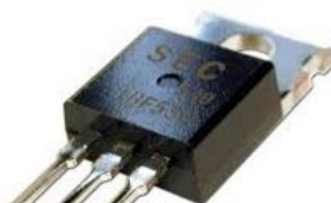
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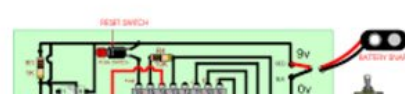
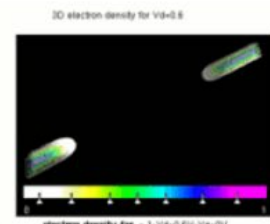
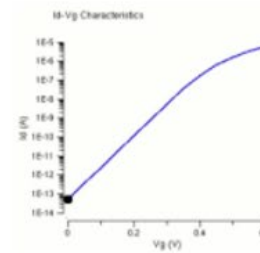
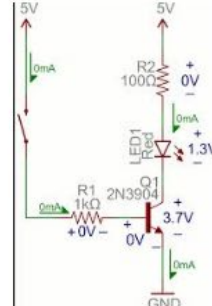
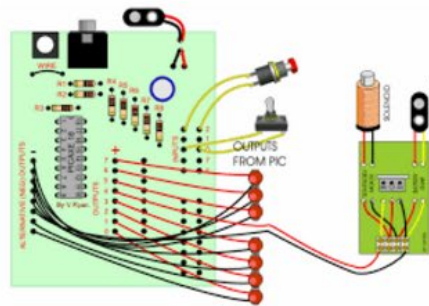
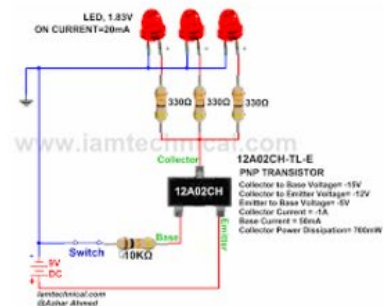
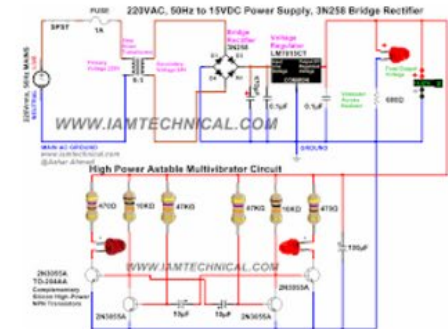
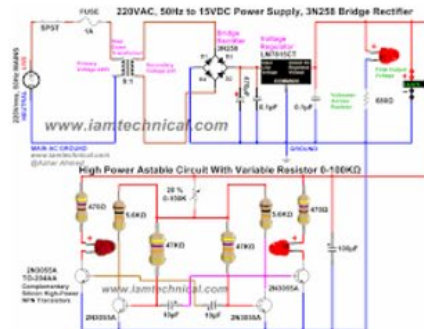
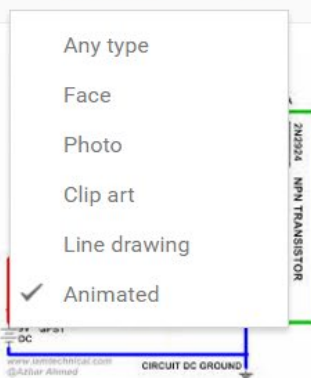
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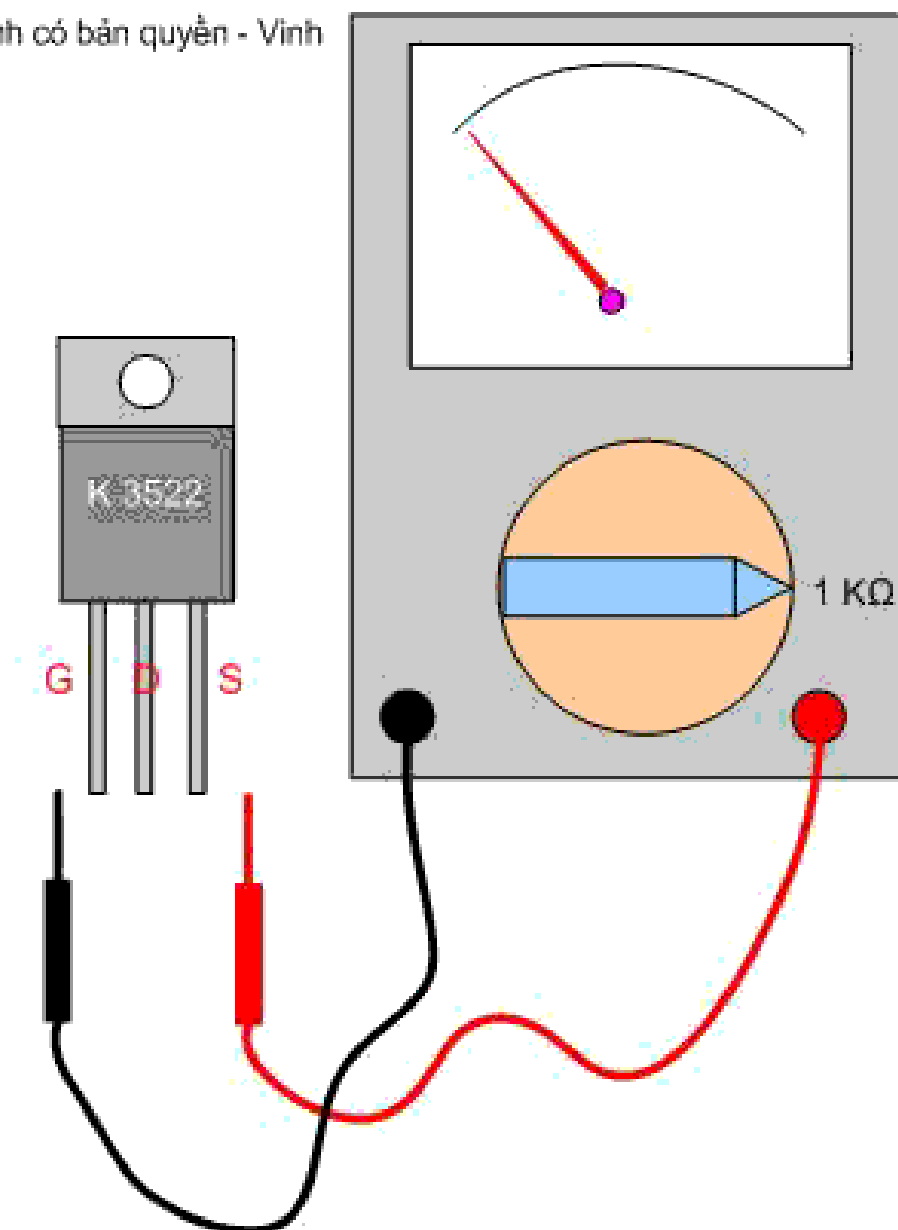
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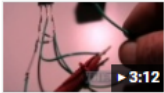
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<https://www.youtube.com/watch?v=GBIVZdgVbWw>
Jan 12, 2015 - Uploaded by TheAstro30
Simple circuit to test the operation of **power transistors** that test fine with a diode test if you don't have a fancy ...

2N3055 Power Transistor Test - YouTube

<https://www.youtube.com/watch?v=qnVGwfc8CB4>
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I show how to test a 2N3055 **power transistor** using a Digital Multimeter.

test npn power transistor - YouTube

<https://www.youtube.com/watch?v=LWox4KfTngw>
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Field-Effect and Bipolar Power Transistor Physics - Page 103



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Adolph Blicher - 2012 - [Preview](#) - [More editions](#)

CHAPTER 6 **POWER TRANSISTOR** STRUCTURES. BIPOLAR TRANSISTOR MODELS Bipolar **power transistors** found applications as low and high frequency amplifiers and oscillators and as power switching devices. One can find them in a ...

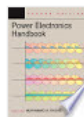
Power-transistor and TTL integrated-circuit applications



<https://books.google.com/books?id=LzNTAAAMAAJ>

Bryan Norris, [Texas Instruments Limited](#). [Applications Laboratory](#) - 1977 - Snippet view

Power Electronics Handbook: Devices, Circuits and Applications



<https://books.google.com/books?isbn=0080467652>

Muhammad H. Rashid - 2010 - [Preview](#) - [More editions](#)

Power. Bipolar. **Transistors**. Marcelo Godoy Simoes, Ph.D. Engineering Division, Colorado School of Mines, Golden, Colorado ... The first **transistor** was discovered in 1948 by a team of physicists at the Bell Telephone Laboratories and soon ...

Lateral Power Transistors in Integrated Circuits - Page 75

Advanced searching

The image shows a Google search interface. The search bar contains the text "power transistor". Below the search bar, the "All" tab is selected, and the search results show "About 28,200,000 results (0.32 seconds)". A snippet of a Wikipedia article is visible, describing a transistor. A dropdown menu is open from the "Settings" link, showing options: "Search settings", "Languages", "Turn on SafeSearch", "Advanced search" (which is highlighted), "History", and "Search help".

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A **transistor** is a semiconductor device used to amplify or switch electrical signals and electrical **power**. It is composed of semiconductor material used with three terminals for connection to an external circuit.

[Transistor - Wikipedia](https://en.wikipedia.org/wiki/Transistor)
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<https://www.onsemi.com/pub/Collateral/2N5684-D.PDF> ▼

2N5684 (PNP), 2N5686 (NPN). High-Current. Complementary Silicon. **Power Transistors**. These packages are designed for use in high-**power** amplifier and.

[PDF] 2N5191 - Silicon NPN Power Transistors - ON Semiconductor

<https://www.onsemi.com/pub/Collateral/2N5191-D.PDF> ▼

Silicon NPN **power transistors** are for use in **power** amplifier and switching circuits – excellent safe area limits. Complement to PNP. 2N5194, 2N5195. Features.

[PDF] LP395 Ultra Reliable Power Transistor (Rev. C) - Texas Instruments

www.ti.com/lit/ds/symlink/lp395.pdf ▼

LP395 Ultra Reliable **Power Transistor**. Check for Samples: LP395. The LP395 is easy to use and only a few precautions. 1FEATURES need be observed.

[PDF] High voltage fast-switching NPN power transistor - STMicroelectronics

www.st.com/resource/en/datasheet/buf420aw.pdf ▼

NPN **POWER TRANSISTOR** s. STMicroelectronics PREFERRED. SALESTYPE s. HIGH VOLTAGE CAPABILITY s. VERY HIGH SWITCHING SPEED s. MINIMUM ...

[PDF] Bipolar Power Transistor Selection Guide

www.mouser.com/catalog/supplier/library/pdf/FairchildBipolarPower.pdf ▼

www.fairchildsemi.com. Bipolar **Power Transistor**. Selection Guide. January 2003. Table of Contents. Product page. General Purpose **Transistors**. Horizontal ...

[PDF] Power Transistor Specification List - Creatron Inc

<https://www.creatroninc.com/upload/Power%20Transistor%20List.pdf> ▼

Power Transistor Specification List. 2N3055. NPN. 60V. 14A. 20 to 70. 155W. 2N6036. PNP, Darlington. -80V. -4A. 750 to 15000. 40W. 2N6039. NPN, Darlington.

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The DOAJ Team is expanding!

I am very happy to announce that the DOAJ Team has expanded. We welcome to the team Ilaria Flava, based in Rome, Italy, and Clara Armengou, based in Barcelona, Spain. Ilaria joins us as Managing Editor and will assist with the monitoring of our editorial groups, the processing of applications and communicating final decisions to [...] [Read More...](#)

Fri, 07 Oct 2016 at 09:52

Open Access Journals Strategy in Algeria

This is a guest post by Kamel Belhamel, DOAJ Ambassador, North Africa. The importance of Open Access (OA) was recently recognised by Algerian scientists, libraries and publishers. Until recently, Algerian researchers preferred to publish their papers in European journals with an impact factor, in order to achieve certain personal goals (e.g. better career opportunities and [...]) [Read More...](#)

Thu, 06 Oct 2016 at 15:49

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Gilmar Arduino Bettio Marodin, Giovambattista Sorrenti, Adamo Domenico Rombolá, André Zambam De Pieri, Enrico Barca Semina: Ciências Agrárias. 2012;33(2):639-654 DOI 10.5433/1679-0359.2012v33n2p639

[Abstract](#) | [Full Text](#)



[Voice-Controlled Artificial Handspeak System](#)

Carlo Fonda, Livio Tenze, Enrique Canessa

International Journal of Artificial Intelligence & Applications. 2014;5(1):107-112 DOI 10.5121/ijai.2014.5108

[Abstract](#) | [Full Text](#)



[Wearable sensors network for health monitoring using e-Health platform](#)

I. Orha, S. Oniga

Carpathian Journal of Electronic and Computer Engineering. 2014;7(1):25-29

[Abstract](#) | [Full Text](#)



[Gesture Controlled Robot using Image Processing](#)

Harish Kumar Kaura, Vipul Honrao, Sayali Patil, Pravish Shetty

International Journal of Advanced Research in Artificial Intelligence. 2013;2(5):69-77 DOI 10.14569/IJARAI.2013.020511

[Abstract](#) | [Full Text](#)



[Development of an Internet of Things based Electricity Load Management System](#)

Dr. D. O. Dike, R.E. Ogu, Dr. L. O. Uzoechi, I. A. Ezenugu

American Journal of Engineering Research. 2016;5(8):199-205

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Importância da vitamina B12 na avaliação clínica de
clinical evaluation of elderly patient

Futterleib, Alexandre, Cherubini, Karen

Scientia Medica. 2005;15(1):74-78

[Abstract](#) | [Full Text](#)



Hydrostatic Pressure Affects In Vitro Maturation of

Isac Karimi, Ali Amini, Mehri Azadbakht, Zahra Rashidi

Cell Journal. 2013;15(4):282-293

[Abstract](#) | [Full Text](#)



Yellow and purple nutsedges survey in the southeastern Buenos Aires Province, Argentina

Eyherabide Juan José, Leaden María Inés, Alonso Sara

Pesquisa Agropecuária Brasileira. 2001;36(1):205-209

[Abstract](#) | [Full Text](#)



The influence of catch trials on the consolidation of motor memory in force field adaptation tasks

Anne eFocke, Christian eStockinger, Christina eDiepold, Marco eTaubert, Thorsten eStein

Frontiers in Psychology. 2013;4 DOI 10.3389/fpsyg.2013.00479

[Abstract](#) | [Full Text](#)



Le quartier épiscopal, campagne 2010, Byllis (Albanie)

Nicolas Beaudry, Pascale Chevalier et Skënder Muçaj

Bulletin du Centre d'Études Médiévales d'Auxerre. 2011;(15):91-95 DOI 10.4000/cem.11925

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Wi-Fi is an important threat to human health

Environmental Research, Volume 164, July 2018, Pages 405-416

Martin L. Pall

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Optimization of solution growth and characterization of monomethylurea crystals

Synthetic Metals, Volume 83, Issue 3, December 1996, Pages 209-211

M. Zha, M. Arduino, G. Zuccalli, L. Zanotti, ... C. Razzetti

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M. Cesari, C. Franchi, L. Cortesi, A. Nobili, ... the REPOSI collaborators

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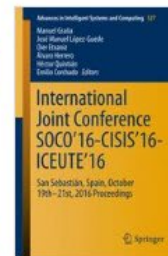


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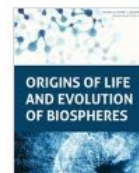
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Rough surfaces with enhanced heat transfer for electronics cooling by direct metal laser sintering



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ABSTRACT

Experimental evidences are reported on the potential of direct metal laser sintering (DMLS) in manufacturing flat and finned heat sinks with a remarkably enhanced convective heat transfer coefficient, taking advantage of artificial roughness in fully turbulent regime. To the best of our knowledge, this is the first study where artificial roughness by DMLS is investigated in terms of such thermal performances. On rough flat surfaces, we experience a peak of 73% for the convective heat transfer enhancement (63% on average) compared to smooth surfaces. On rough (single) finned surfaces, the best performance is found to be 40% (35% on average) compared to smooth finned surface. These results refer to setups with Reynolds numbers (based on heated edge) within $3500 \leq Re_L \leq 16,500$ (corresponding to $35,000 \leq Re_D \leq 165,000$ in terms of Reynolds number based on hydraulic diameter). Experimental data are obtained by a purposely developed sensor with maximum and mean estimated tolerance intervals of $\pm 7.0\%$ and $\pm 5.4\%$, respectively. Following the idea by Gioia et al. (2006) [48], we propose that heat transfer close to the wall is dominated by eddies with size depending on the roughness dimensions and the viscous (Kolmogorov) length scale. An excellent agreement between the experimental data and the proposed analytical model is finally demonstrated.

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1. Introduction and motivation

Thermal management of the microprocessors used in notebook and desktop computers often relies on chip-attached or adhesively bonded extruded aluminum heat sinks, cooled by remotely located fans [1]. In particular, battery power limitations in notebook computers represent a motivation to keep searching for heat sinks with enhanced performances. Highly efficient heat sinks, with reduced thermal resistances, are required also by high-end commercial workstations and servers. Many details about the thermal management of electronic devices and practical issues associated with the efficient packaging are reported in Refs. [2,3]. Even though water-based two-phase cooling systems are known to ensure remarkably high heat fluxes (two- or three orders of magnitude higher than forced air systems), it is difficult to imagine a widespread use of such a technology in notebook computers, which will remain dominated by forced air convection cooling systems reasonably for long time. However, in the next-generation electronics devices, thermal

performances of the air-cooled heat sinks must be further improved due to a steadily increasing power density, which makes the thermal management a great challenge still to be faced in the next future [4].

Forced air heat transfer enhancement has been extensively explored and many augmentation techniques have been already proposed [5], including plane fins [6,7], pin fins [8–10], dimpled surfaces [11–13], surfaces with arrays of protrusions [14,15], metal foams [16], and artificial surface roughness [17]. By *artificial surface roughness*, we mean any surface patterning with enough regularity and purposely designed in order to enhance heat transfer. For instance, in such a category, we may include ribs [18–20] and, more recently, (shark-skin-like) scale roughened surfaces [21,22]. The resulting heat transfer enhancement of the scale roughened surface is surprisingly good compared to rib roughened and dimpled surfaces [23]. This proves that there is still room for improving the optimal design of artificial surface roughness. To this respect, an interesting possibility consists in adopting a multi-scale strategy, where pin micro-fins are placed on standard plate fins. Recently, Authors in Ref. [24] showed that pin fins of five different cross-section shapes in channels of plate-fin heat sinks cause enhancement in the heat transfer. Short pin fins, on surfaces

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